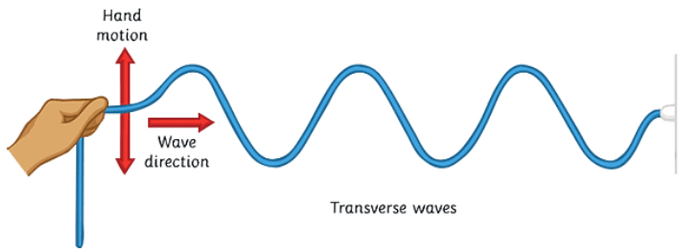
Knowledge Organiser
Year 8: Waves – Wave Properties

What is a wave?
- A wave is a disturbance which transfers energy
- Waves may vibrate matter but they do not move it from its original location
- Waves come in two types; transverse and longitudinal

Diagram which shows the **action of a transverse wave** on a rope



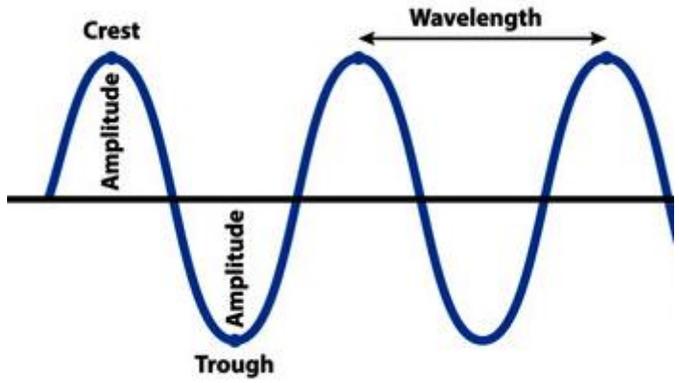
Transverse waves?
- In a transverse wave the vibrations are perpendicular (at right angles) to the direction of wave travel
- Examples of transverse waves include: ripples on the surface of water, electromagnetic waves

Diagram which shows the **action of a longitudinal wave**



Longitudinal waves?
- In a **longitudinal wave** the vibrations are **parallel** to the direction of the wave travel
- Examples of longitudinal waves include: sound waves, seismic waves

Diagram showing the labelled features of a transverse wave



What are the features of a wave?
- Amplitude is the distance from the mid-point to the peak/trough
- It is the maximum disturbance of a particle caused by the vibration
- Wavelength is the distance taken for a complete wave
- Wavelength is usually measured from peak to peak or trough to trough

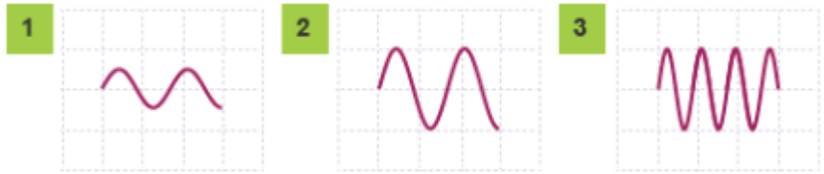
What about frequency?
- Frequency is the number of complete waves which are produced **each second**
- The unit of frequency is **hertz (Hz)**
- The **time period** of a wave is the time taken for one complete wavelength
- It is calculated as 1/frequency

	Definition	Symbol	Units
Wavelength	Distance from peak to peak	λ	Meters (m)
Frequency	Number of waves per second	f	Hertz (Hz)
Time period	Time taken for a complete wave	T	Seconds (s)
Wave speed	Speed a wave travels at	v	Meters per second (m/s)

How do we change the pitch of a sound?
- Pitch is how high or low a sound is
- It is determined by the frequency of the wave
- High frequencies are high pitched

How do we change the volume of a sound?
- Volume is a measure of loudness
- It is determined by the amplitude of the wave
- A high amplitude will sound loud

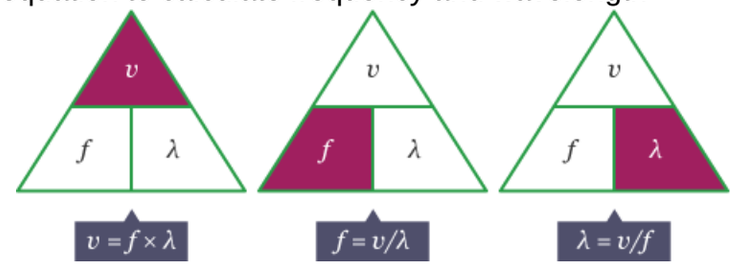
Diagram showing the wave form of;
1. quiet, low pitch sound, 2. loud, low pitch sound and 3. loud, high pitch sound



$$v = f \times \lambda$$

Wave speed equation:
Wave speed = Frequency x Wavelength

Diagrams showing how to **rearrange** the wave speed equation to calculate frequency and wavelength



RAG Yourself!

Learning Outcomes;	RAG
Describe the difference between transverse and longitudinal waves	
Define parallel and perpendicular	
Label a diagram with wavelength and amplitude	
Define frequency and time period of a wave	
Describe how to change the pitch and volume of a sound in terms of wave shape	
Rearrange the wave speed equation to calculate frequency and wavelength	

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Knowledge Organiser
Year 8: Waves – Wave Properties

What is a wave?

Diagram which shows the **action of a transverse wave** on a rope

Transverse waves?

Diagram which shows the **action of a longitudinal wave**

Longitudinal waves?

Diagram showing the labelled features of a transverse wave

What are the features of a wave?

What about frequency?

	Definition	Symbol	Units
Wavelength			
Frequency			
Time period			
Wave speed			

How do we change the pitch of a sound?

How do we change the volume of a sound?

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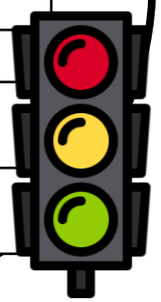
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Wave speed equation:

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